

Design Document

For

Online Plant Purchasing Nursery system

Version 1.0

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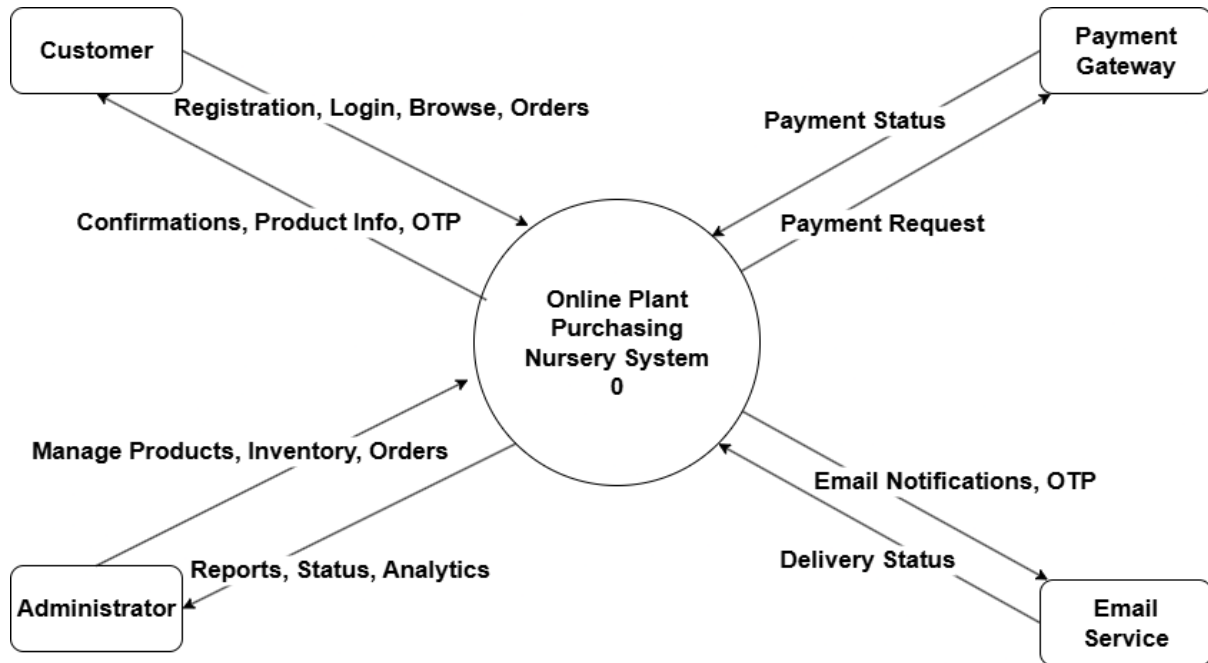
Department Of Information Technology, NITK Surathkal

TABLE OF CONTENTS

1. DATA FLOW DIAGRAM MODEL	3
1.1 Level-0: Context Diagram	3
1.2 Level-1	3
1.3 Level-2	4
1.4 Level-3	8
2. STRUCTURE CHART	10
3. CORE MODULES	11
3.1 User Management	11
3.2 Product Catalog Management	11
3.3 Shopping Cart Management	11
3.4 Order Processing	11
3.5 Inventory Management	12
3.6 Other Managements	12
3.7 Customer Support	12
4. DATA DICTIONARY	12
4.1 External Entities	12
4.2 Key Data Flows	13
Level 2 Data Flows	
User Management System	13
Product Catalog Management System	15
Inventory Management System	17
Order Processing System	18
Shopping Cart Management System	20
Customer Support System	22
Other Management Systems	24
Level 3 Data Flows	
JWT Token Generation Process	25
Device Session Management	27
Search and Filter Processing	27
Cart Contents Validation Process	28
Order Tracking Process	31

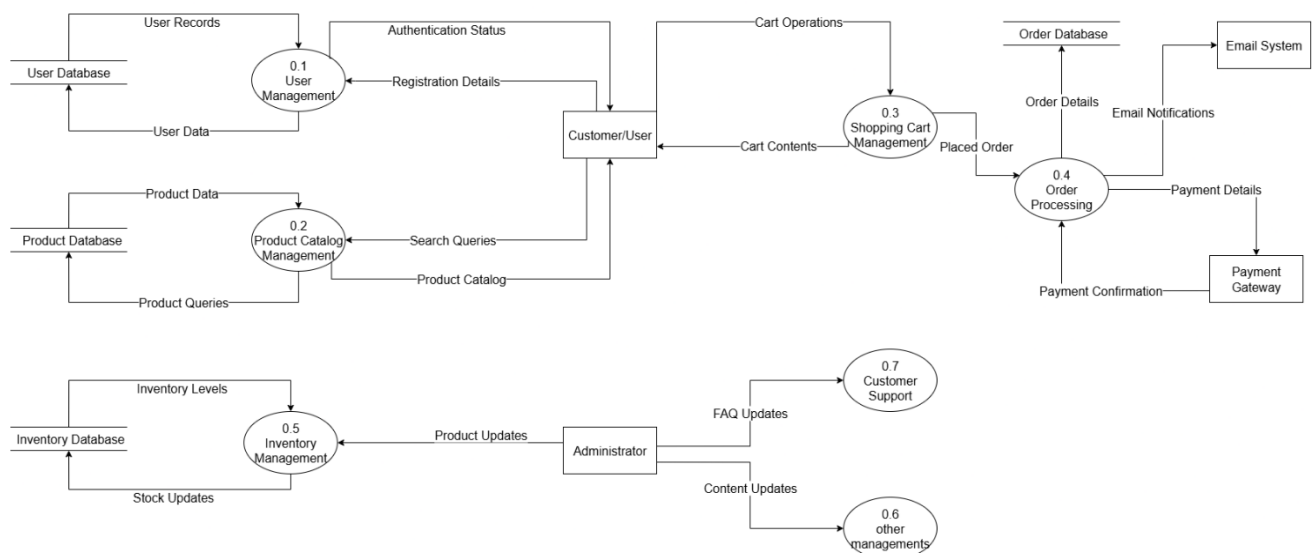
1.DATA FLOW DIAGRAM MODEL

LEVEL-0: CONTEXT DIAGRAM

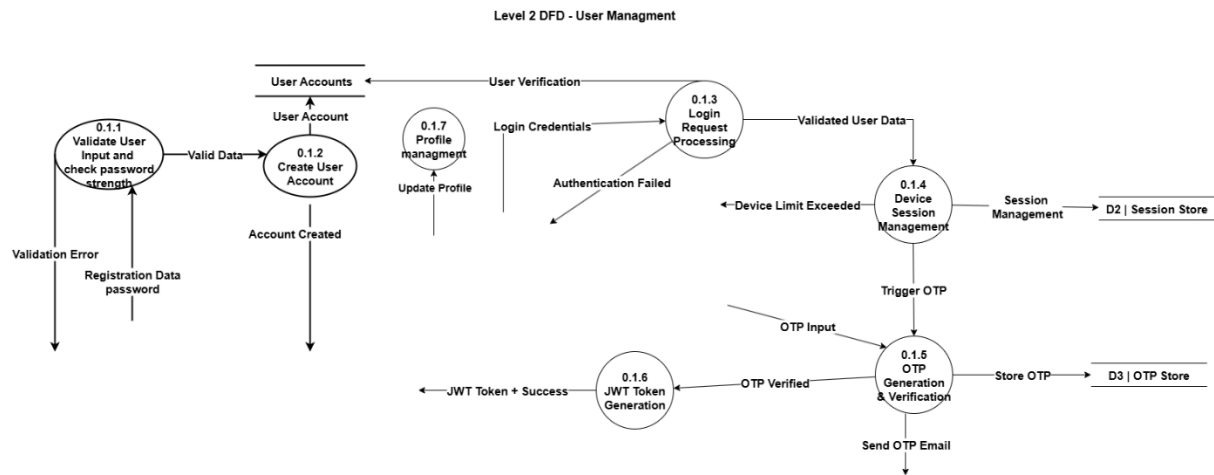


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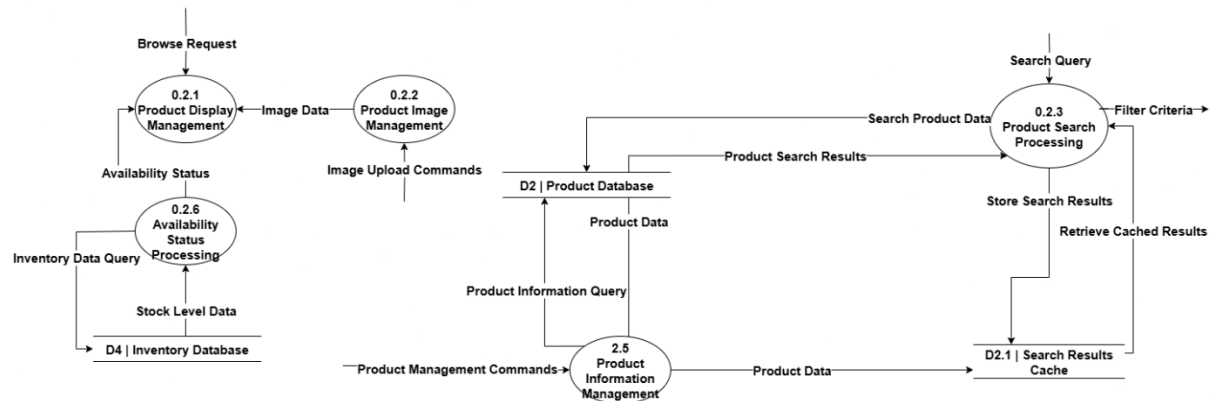
Level 1 Data Flow Diagram



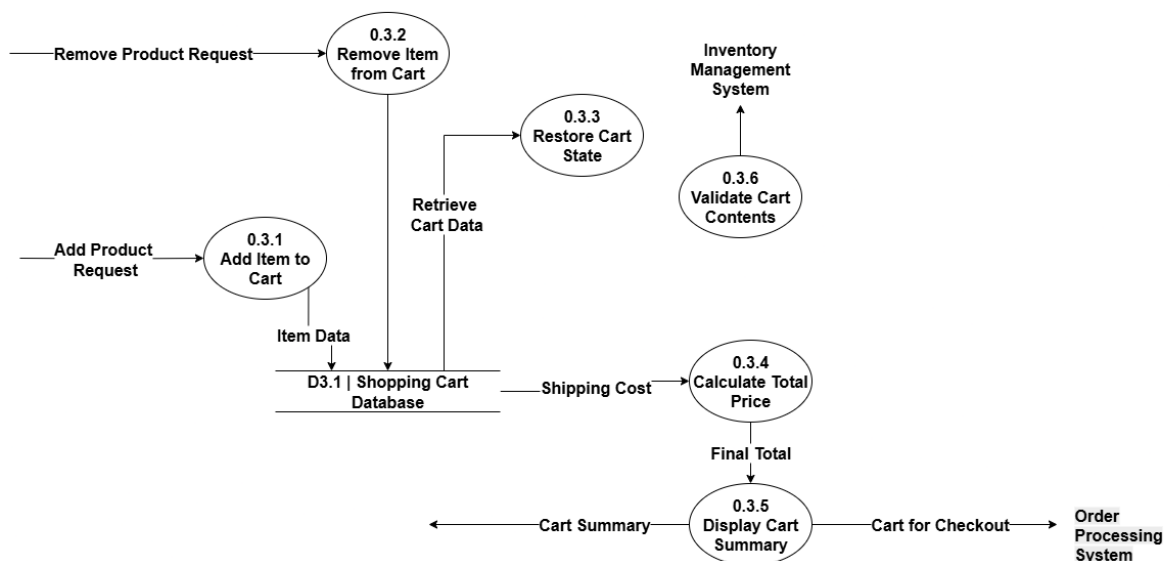
LEVEL-2:



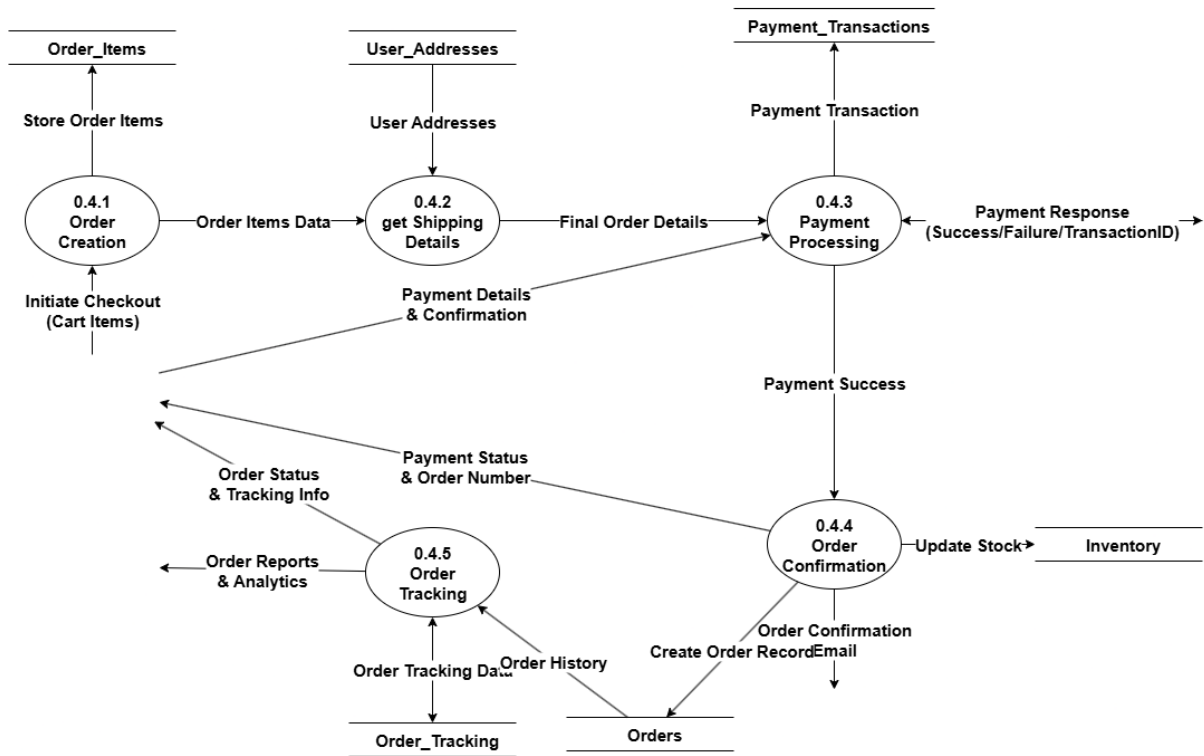
Level 2 DFD for Product Catalog Management



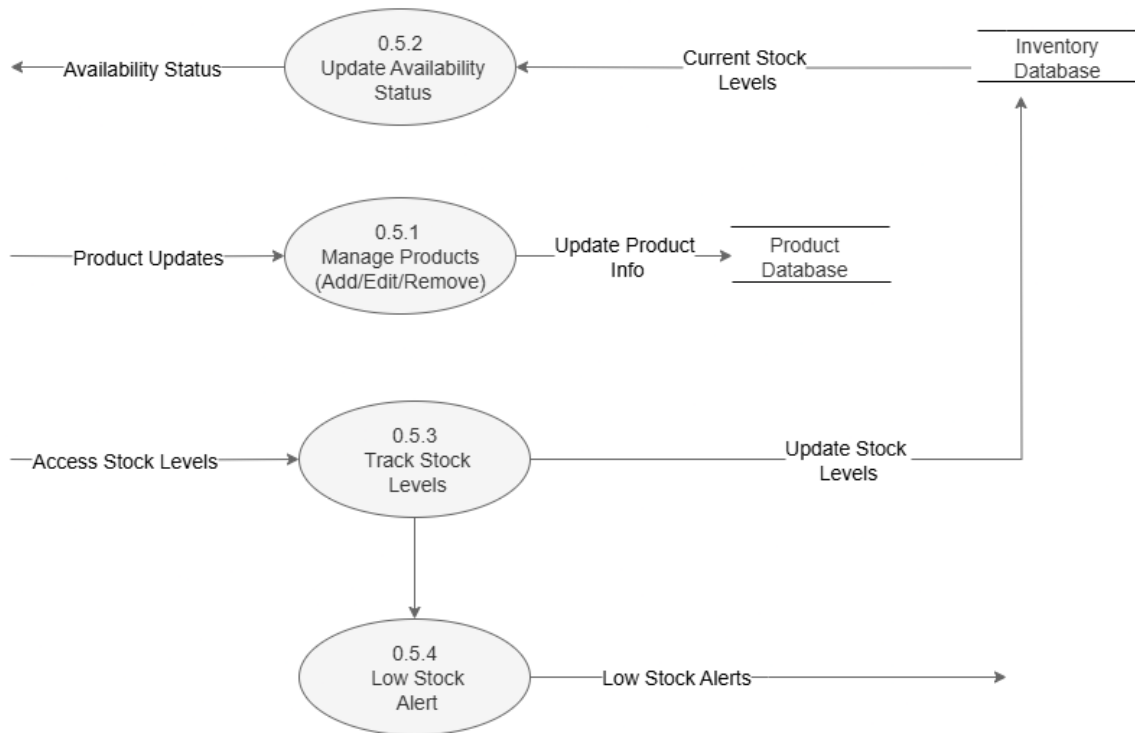
Level 2 DFD for Shopping Cart Management



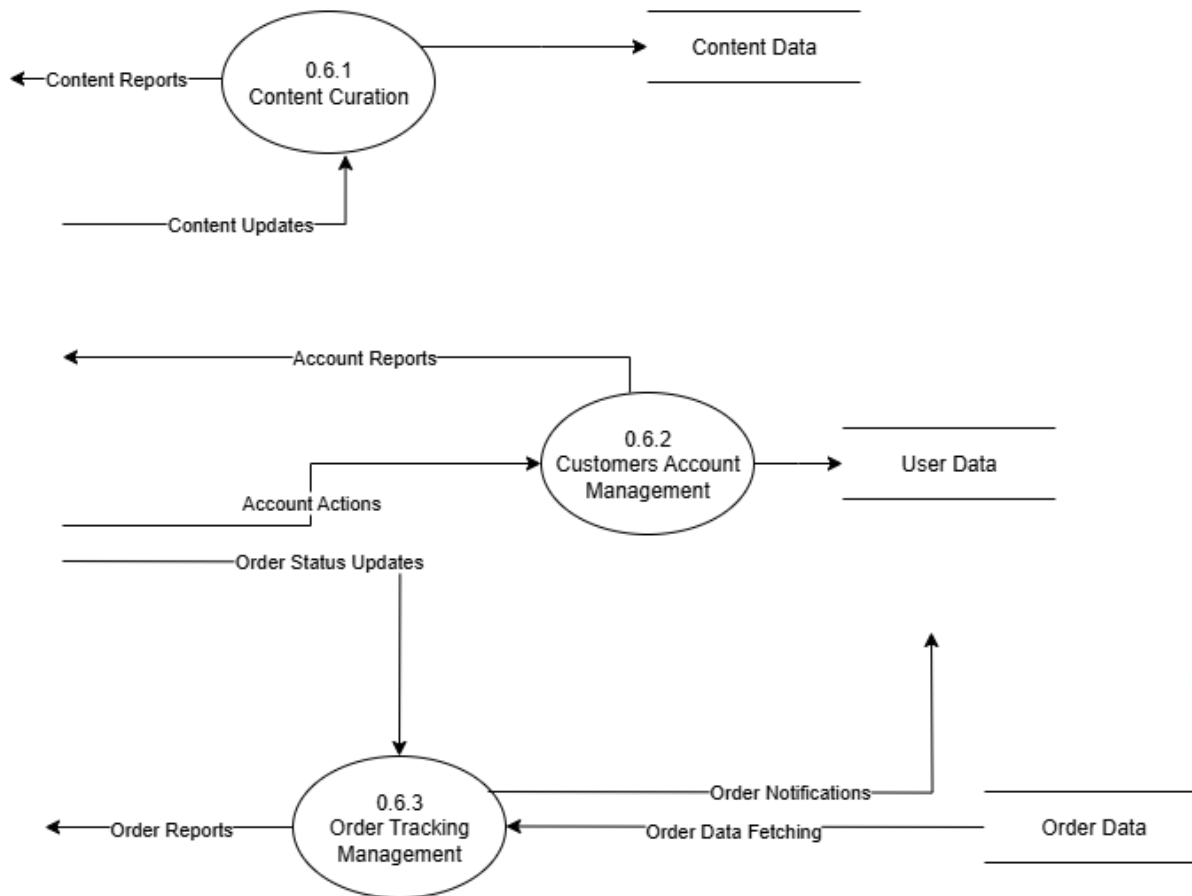
Level 2 DFD - Process 4.0 Order Processing



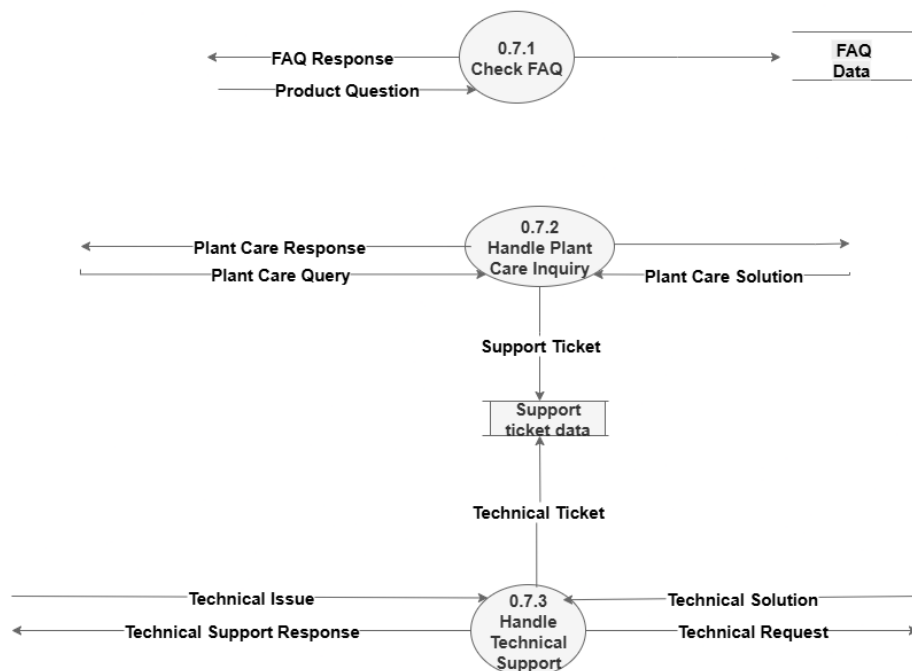
Level 2 DFD for Inventory Management



Level 2 DFD for Other Managements

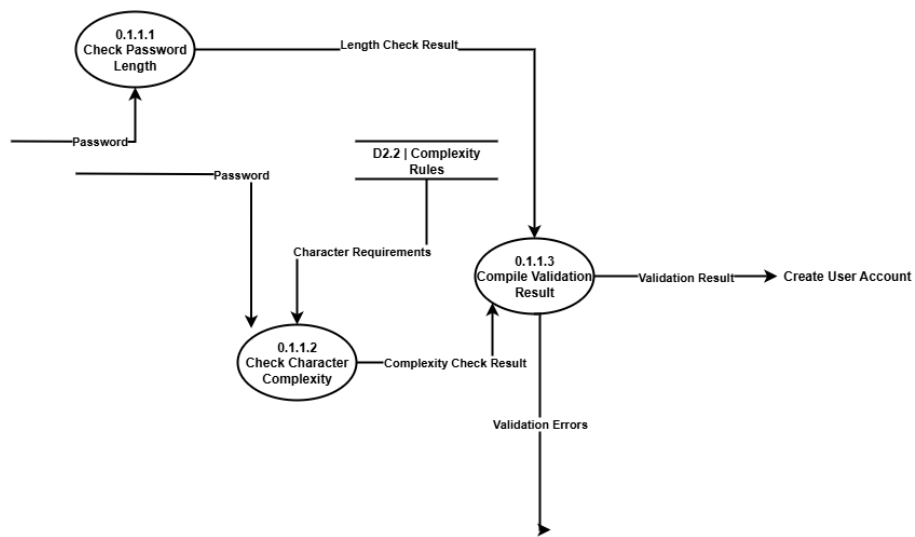


Level 2 DFD for Customer Support

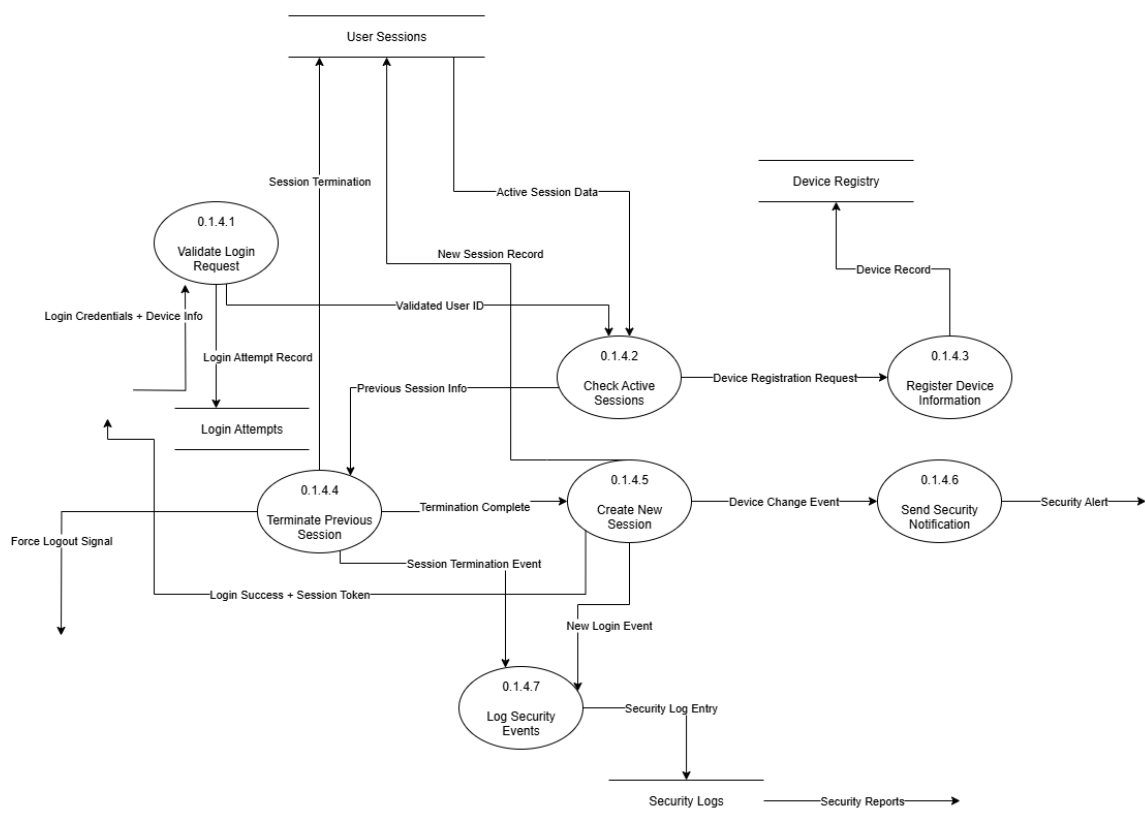


LEVEL -3:

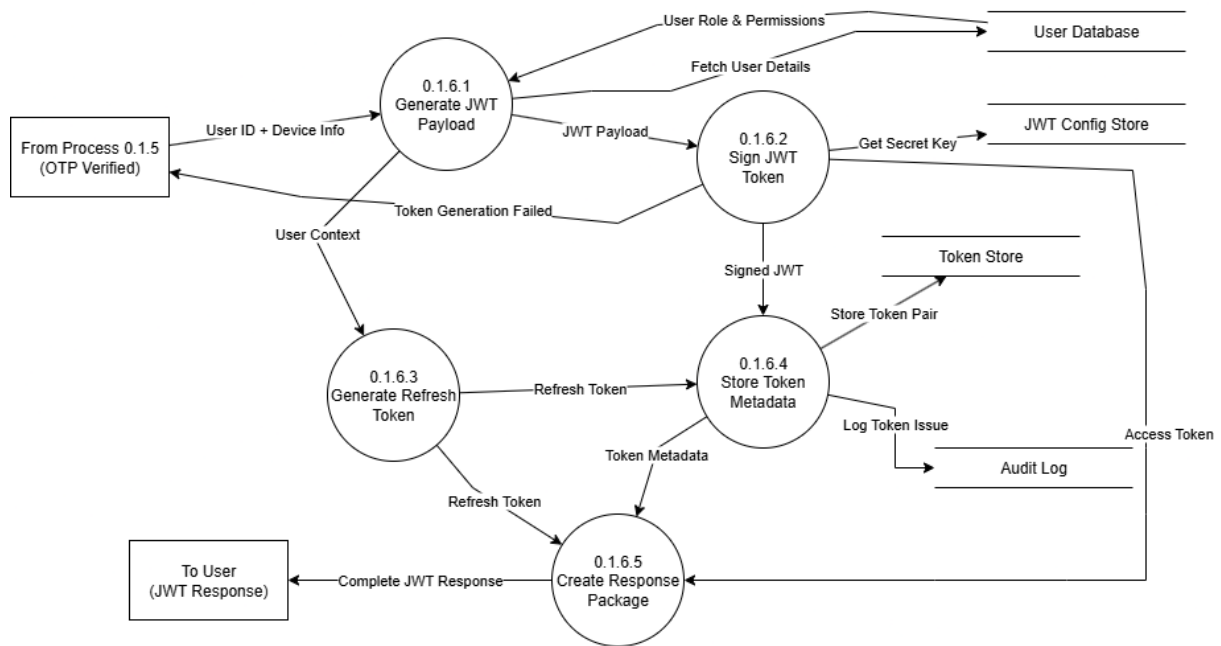
Level 3 DFD - Password Strength Validation



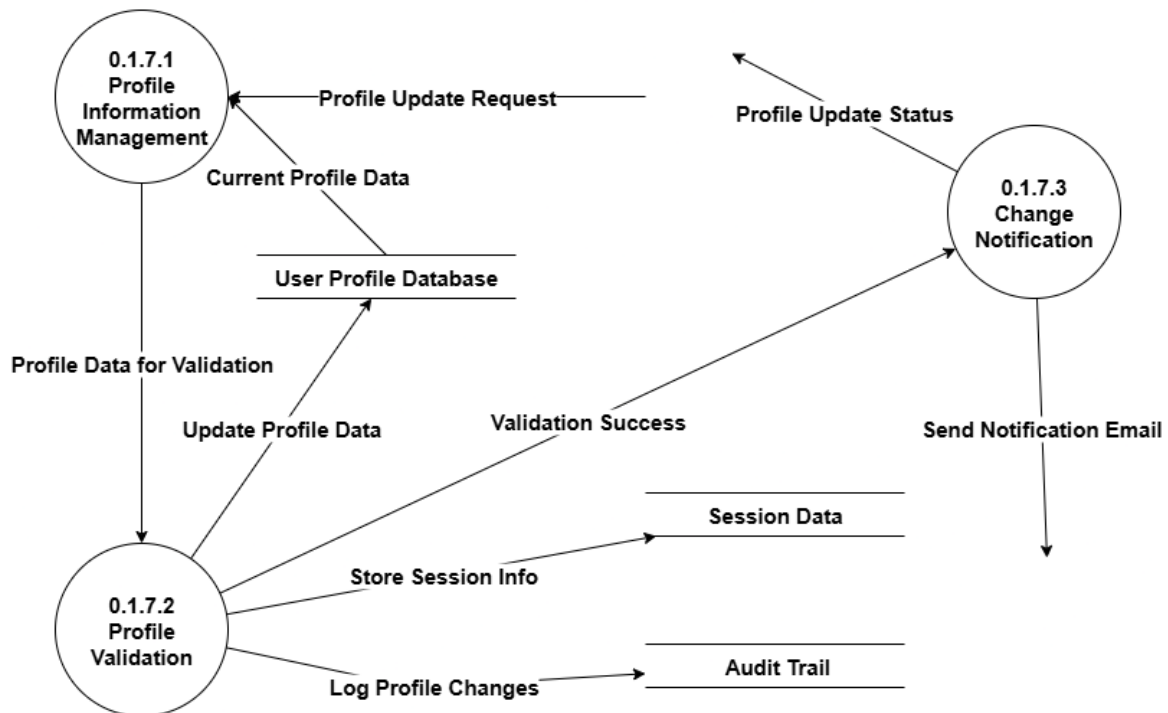
Level 3 DFD - Device Session Management



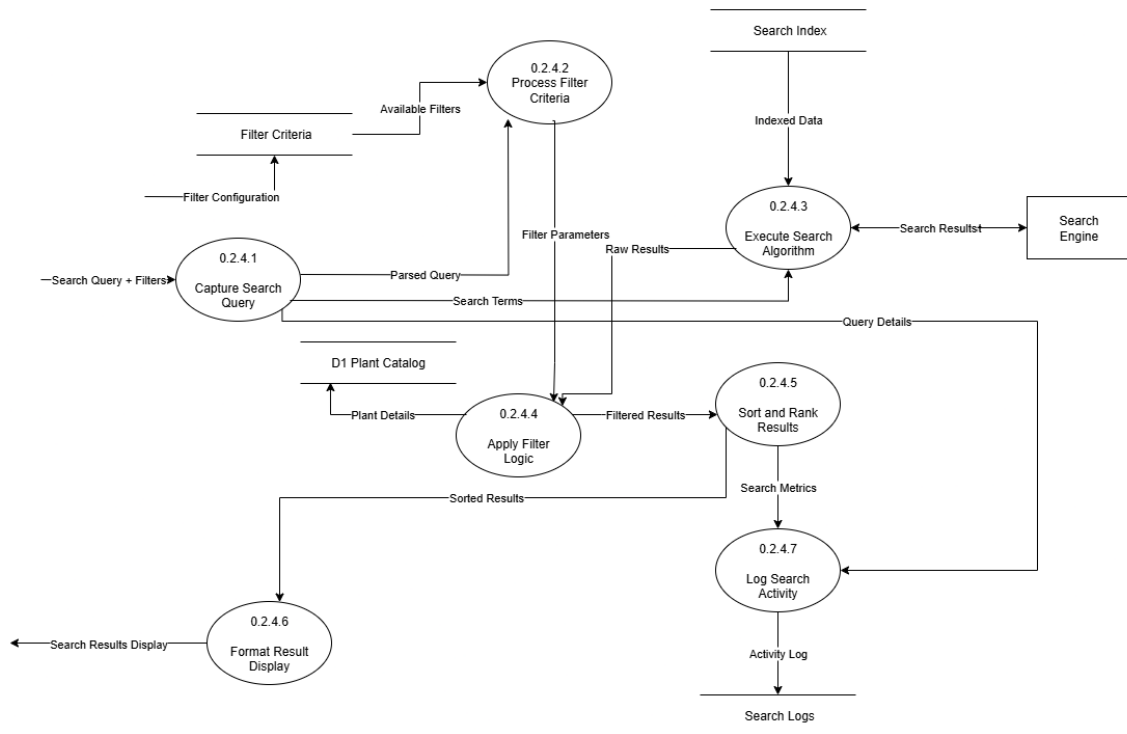
Level 3 DFD - JWT Token Generation Process



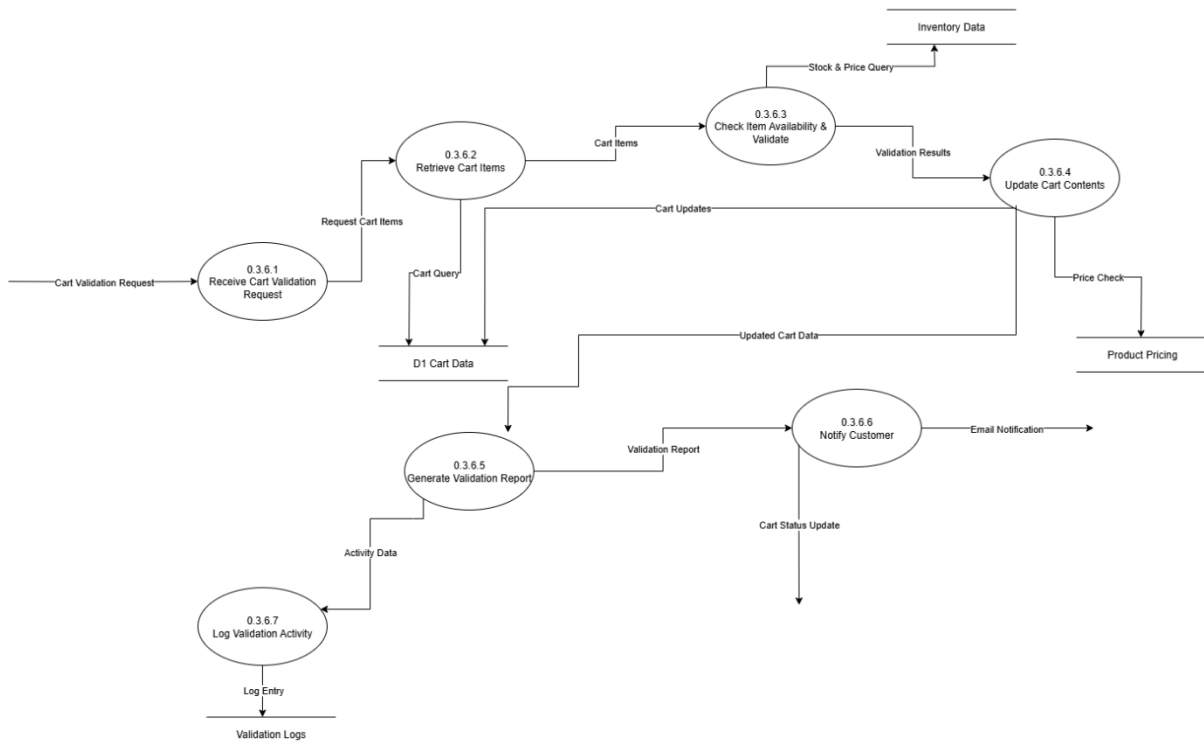
Level 3 DFD - Profile Management System (0.1.7)



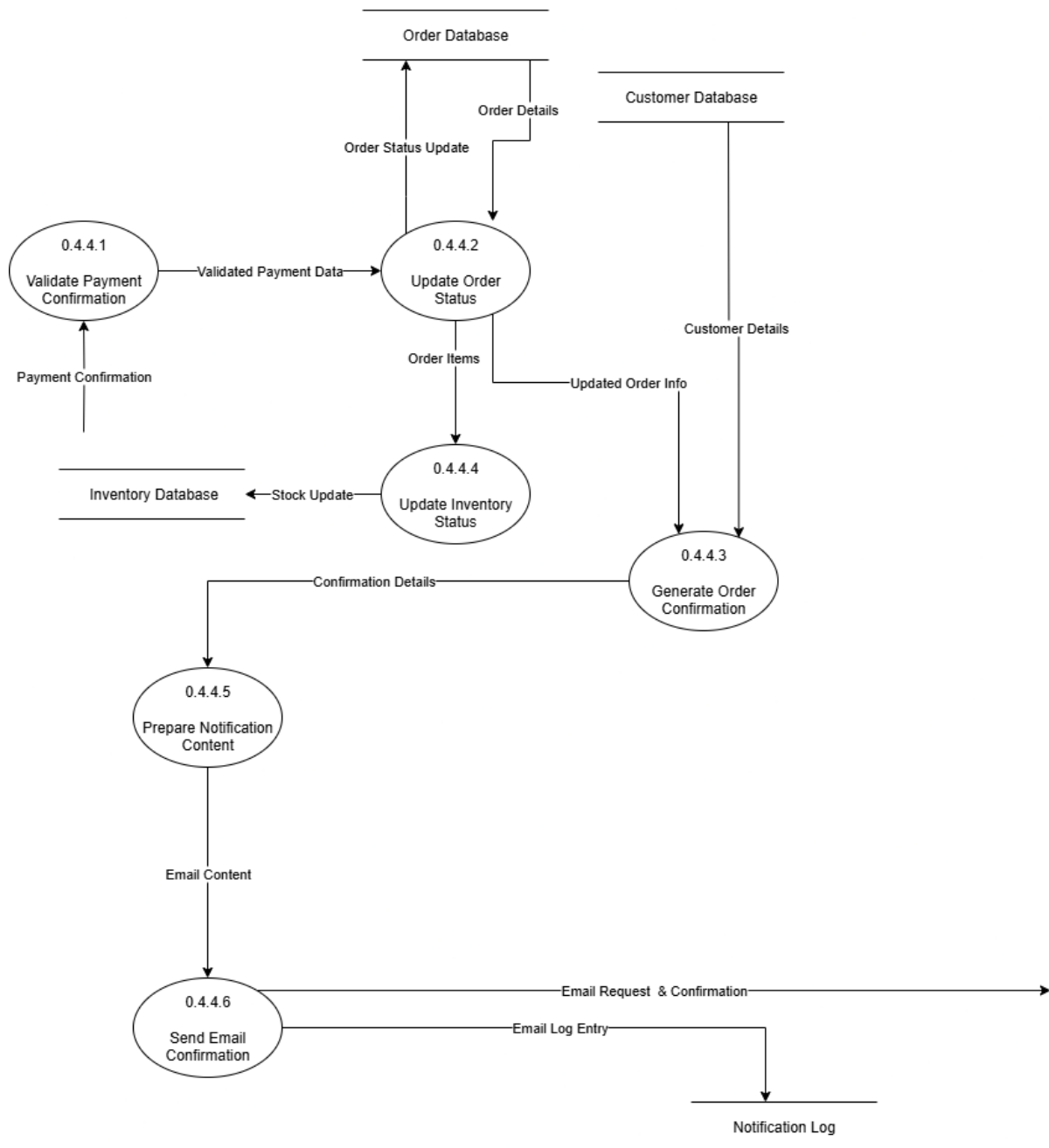
Level 3 DFD - Search and Filter Processing



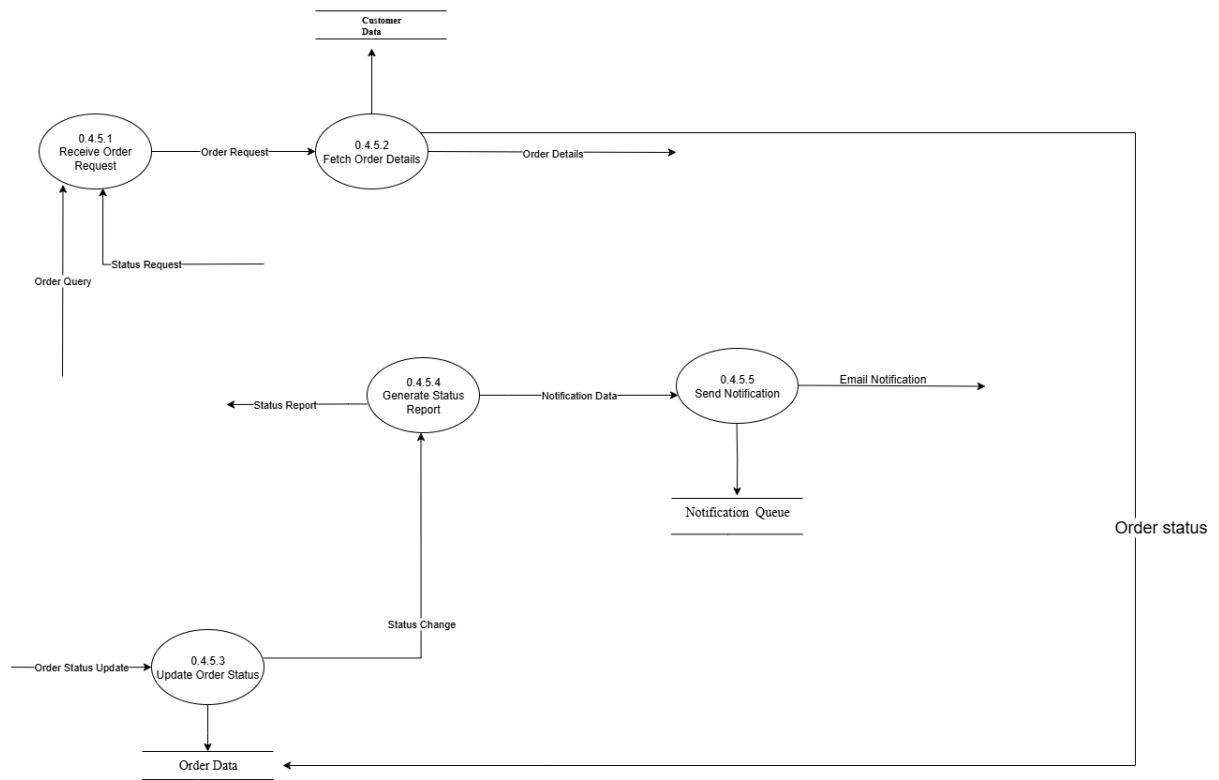
Level 3 DFD - Validate Cart Details



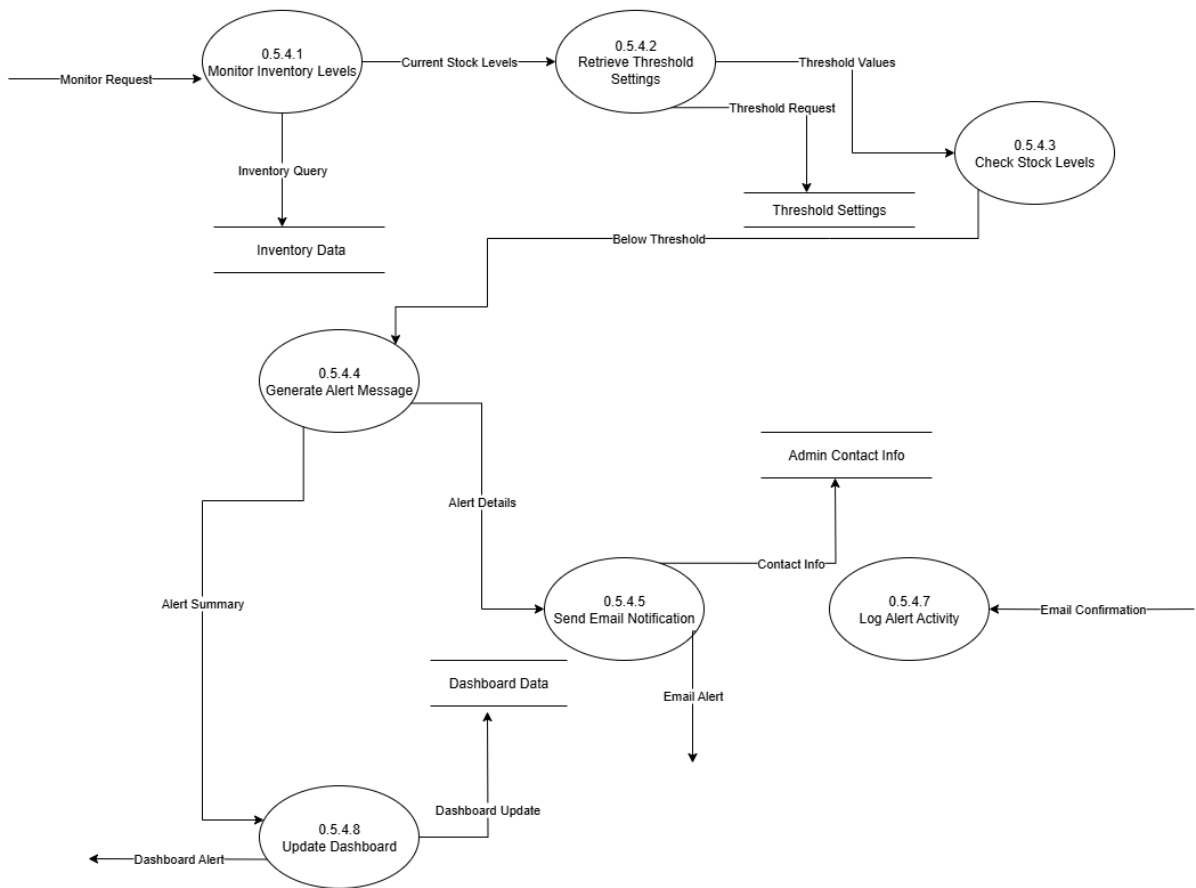
Level 3 DFD - Order Confirmation



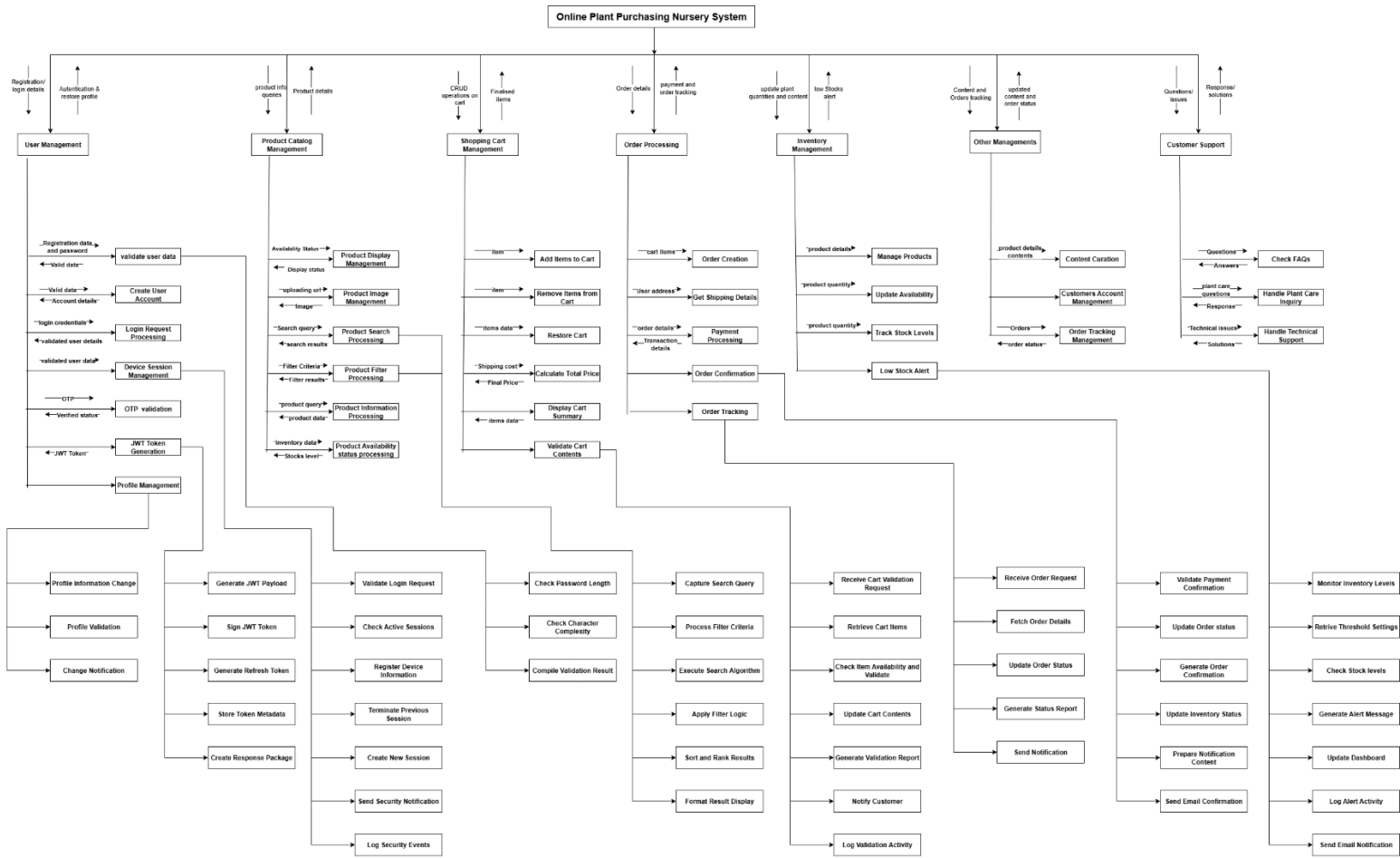
Level 3 DFD - Order Tracking



Level 3 DFD - Low Stock Alerts



2.STRUCTURE CHART



*Didn't include data details for level 3, because it started looking clumsy

3. Core Modules

This section describes the main modules of the Online Plant Purchasing Nursery system. Each module has been assigned to a designer for detailed implementation, and the functions under each module are listed below.

3.1 User Management

- Handles user registration, login, OAuth integration, password reset, session management, and profile data management.
- Designer: K Akhilesh
- Functions: Register User, Login using Email/Password, OAuth Callback & Link Accounts, Password Reset, Session Management

3.2 Product Catalog Management

- Manages the plant catalog through display and operations such as search and filtering.
- Designer: Charan Gowda CM
- Functions: Display management, image management, search and filter processing, and availability status

3.3 Shopping Cart Management

- Allows users to add, remove, and restore items, then validates cart contents to calculate the total price.
- Designer: M Lakshya
- Functions: Add items, remove items, restore cart, display cart, validate cart contents, and calculate price

3.4 Order Processing

- Creates and places orders with third-party payment method integration.
- Designer: M Lakshya
- Functions: Order creation, get shipping address, order confirmation, order tracking

3.5 Inventory Management

- Manages products by tracking and updating product quantities.
- Designer: Charan Gowda CM
- Functions: Manage products, update availability, track stock levels, low stock alert

3.6 Other Managements

- Manages content and handles order updates.
- Designer: Charan Gowda CM
- Functions: Content curation, customer account management, order updates

3.7 Customer Support

- Assists in resolving customer issues including technical glitches, plant care inquiries, and order-related problems.
- Designer: K Akhilesh
- Functions: FAQs, handle plant care enquiries, handle technical support

4. Data Dictionary

4.1 External Entities

Entity Name	Description
Customer/User	End users who browse products, register accounts, make purchases, and access plant care information
Administrator	System administrators who manage products, users, content, and system operations
Payment Gateway	External payment processing service that handles secure payment transactions
Email System	External email service for sending notifications, confirmations, and alerts
Inventory Management System	External system that provides real-time stock level data and availability information

4.2 Key Data Flows

Level 2: User Management System

Data Flow	Data Elements	Type	Description
User Accounts	User_ID, Username, Email, Account_Status, Registration_Date, Last_Login	Structured	User account information and status

User Verification	User_ID, Verification_Type, Verification_Code, Expiry_Time, Attempt_Count	Structured	User identity verification data
User Data	User_Profile, Personal_Info, Preferences, Contact_Details, Account_Settings	Structured	Complete user profile and personal data
Login Credentials	Username, Password_Hash, Login_Timestamp, IP_Address, Device_Info	Structured	User authentication credentials
Registration Data	New_User_Data, Email, Password, Personal_Info, Terms_Acceptance	Structured	New user registration information
Account Created	User_ID, Account_Status, Creation_Timestamp, Welcome_Message	Structured	Successful account creation confirmation
Validation Error	Error_Code, Error_Message, Field_Name, Validation_Rule, User_Input	Structured	Account validation failure information
New User	User_Record, Profile_Data, Account_Credentials, Initial_Settings	Structured	Complete new user record for storage

User Database	User_ID, Account_Data, Profile_Info, Security_Settings, Activity_Log	Structured	User information stored in database
Authentication Failed	Failed_Attempt, User_ID, Error_Type, IP_Address, Timestamp, Lockout_Status	Structured	Failed authentication attempt data
Device Limit Exceeded	User_ID, Current_Device_Count, Device_List, Limit_Threshold, Action_Required	Structured	Device limit violation notification
Trigger OTP	User_ID, Phone_Number, OTP_Type, Generation_Time, Delivery_Method	Structured	OTP generation trigger data
JWT Token + Success	Access_Token, Refresh_Token, User_ID, Token_Expiry, Success_Status	Structured	Successful authentication token response
JWT Token Generation	Token_Payload, User_Claims, Expiry_Time, Signature_Algorithm	Structured	JWT token generation data
OTP Input	User_ID, OTP_Code, Input_Timestamp, Verification_Attempt	Structured	User-entered OTP for verification

Level 2: Product Catalog Management System

Data Flow	Data Elements	Type	Description
Browse Request	User_ID, Session_ID, Category_ID, Page_Number, Items_Per_Page	Structured	User request to browse and view available products in the catalog
Search Query	Search_Keywords, Category_Filter, Price_Min, Price_Max, Brand_Filter, Sort_Order	Structured	User-initiated search request containing search criteria and keywords
Image Data	Image_ID, Product_ID, Image_File, Image_Format, Image_Size, Alt_Text, Upload_Timestamp	Binary/Structured	Product image files and related visual content data
Availability Status	Product_ID, Stock_Quantity, Availability_Flag, Reorder_Level, Last_Updated	Structured	Current stock availability information for products
Inventory Data Query	Product_ID, Query_Type, Requested_Fields, Query_Timestamp	Structured	Request for current inventory levels and stock information
Stock Level Data	Product_ID, Current_Stock, Reserved_Stock, Available_Stock, Reorder_Point, Last_Restocked	Structured	Current stock quantities and availability data
Image Upload Commands	Admin_ID, Product_ID, Image_File, Image_Type, Upload_Action, Metadata	Binary/Structured	Administrative commands for uploading and managing product images
Product Management Commands	Admin_ID, Product_ID, Action_Type, Product_Data, Timestamp, Validation_Rules	Structured	Administrative commands for creating, updating, and managing product information

Product Information Query	Query_ID, Product_ID, Requested_Fields, Search_Context, Query_Timestamp	Structured	Request for detailed product information during search operations
Search Product Data	Product_ID, Product_Name, Description, Price, Category, Brand, Specifications, Images	Structured	Product data returned in response to search queries
Product Search Results	Query_ID, Result_Count, Product_List, Applied_Filters, Sort_Order, Page_Info	Structured	Formatted search results containing matching products
Product Data	Product_ID, Product_Name, Description, Price, Category_ID, Brand_ID, SKU, Specifications	Structured	Product information being stored or retrieved from database
Filter Criteria	Category_Filters, Price_Range, Brand_Filters, Attribute_Filters, Sort_Criteria, Result_Limit	Structured	Search filters and criteria for refining product results
Store Search Result	Cache_Key, Search_Results, Result_Count, Query_Hash, Cache_Timestamp, Expiry_Time	Structured	Processed and filtered search results for caching
Retrieve Cached Results	Cache_Key, Query_Hash, Last_Access_Time, Cache_Hit_Flag	Structured	Request to retrieve previously cached search results

Level 2: Inventory Management System

Data Flow	Data Elements	Type	Description
Availability Status	Product_ID, Stock_Status, Available_Quantity, Reserved_Quantity, Status_Timestamp	Structured	Current product availability information

Current Stock Levels	Product_ID, Current_Stock, Minimum_Stock, Maximum_Stock, Unit_of_Measure	Structured	Real-time inventory stock level data
Inventory Database	Inventory_ID, Product_ID, Stock_Quantity, Location_ID, Last_Updated, Batch_Number	Structured	Complete inventory record information
Product Updates	Product_ID, Update_Type, New_Stock_Level, Adjustment_Reason, Admin_ID, Timestamp	Structured	Administrative product inventory updates
Manage Products (Add/Edit/Remove)	Product_ID, Product_Action, Product_Details, Stock_Info, Admin_Authorization	Structured	Product management operations data
Update Product Info	Product_ID, Updated_Fields, New_Values, Validation_Status, Change_Log	Structured	Product information update requests
Product Database	Product_ID, Product_Name, Description, Category_ID, Price, Stock_Info, Status	Structured	Master product information storage
Update Stock Levels	Product_ID, Stock_Adjustment, Adjustment_Type, Reason_Code, Reference_Number	Structured	Stock level modification data

Level 2: Order Processing System

Data Flow	Data Elements	Type	Description
Order Items	Order_ID, Product_ID, Quantity, Unit_Price, Product_Name, Subtotal	Structured	Items selected by customer for order processing
User Addresses	User_ID, Address_ID, Street_Address, City, State, Postal_Code, Country, Address_Type	Structured	Customer shipping and billing address information

Store Order Items	Order_ID, Product_List, Order_Timestamp, Customer_ID, Order_Status	Structured	Order items being stored in the system
User Addresses	Address_ID, Full_Address, Address_Validation_Status, Default_Flag	Structured	Processed user address data
Order Items Data	Order_ID, Item_Details, Pricing_Info, Inventory_Check_Flag	Structured	Detailed order item information for processing
Get Order Details	Order_ID, Customer_ID, Query_Type, Requested_Fields	Structured	Request for complete order information
Payment Transactions	Transaction_ID, Order_ID, Payment_Method, Amount, Currency, Transaction_Status	Structured	Payment processing transaction data
Payment Transaction	Payment_ID, Order_ID, Amount, Payment_Gateway, Processing_Status, Timestamp	Structured	Individual payment transaction record
Payment Details & Confirmation	Payment_ID, Confirmation_Number, Transaction_Status, Payment_Method, Receipt_Data	Structured	Payment confirmation and receipt information
Payment Response	Transaction_ID, Payment_Status, Response_Code, Gateway_Response, Error_Message	Structured	Payment gateway response data
Payment Gateway	Gateway_ID, Payment_Request, API_Credentials, Transaction_Data	Structured	Payment processing gateway communication
Payment Success	Order_ID, Payment_ID, Success_Flag, Confirmation_Code, Receipt_URL	Structured	Successful payment notification data

Initiate Checkout (Cart Items)	Cart_ID, User_ID, Item_List, Total_Amount, Checkout_Timestamp	Structured	Cart items ready for checkout process
Order Status & Tracking Info	Order_ID, Order_Status, Tracking_Number, Estimated_Delivery, Status_History	Structured	Order tracking and status information
Order Receipts & Analytics	Receipt_ID, Order_Summary, Analytics_Data, Customer_Metrics, Sales_Data	Structured	Order completion receipts and business analytics
Order Tracking	Tracking_ID, Order_ID, Current_Status, Location_Updates, Delivery_Estimate	Structured	Real-time order tracking information
Update Stock	Product_ID, Quantity_Sold, New_Stock_Level, Stock_Action_Type	Structured	Inventory update after order processing
Inventory	Product_ID, Available_Stock, Reserved_Stock, Reorder_Level, Stock_Status	Structured	Current inventory levels and status
Order Confirmation	Order_ID, Confirmation_Number, Order_Summary, Delivery_Info, Customer_Details	Structured	Final order confirmation data
Create Order (Receipt-less)	Order_Data, Customer_Info, Payment_Status, Processing_Flag	Structured	Order creation without receipt generation
Email Service	Email_Template, Recipient_Info, Order_Details, Notification_Type	Structured	Email notification service data

Level 2: Shopping Cart Management System

Data Flow	Data Elements	Type	Description
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Remove Product Request	Cart_ID, Product_ID, User_ID, Remove_Action_Type, Timestamp	Structured	Customer request to remove item from shopping cart
Add Product Request	Product_ID, Quantity, User_ID, Cart_ID, Price, Product_Details	Structured	Customer request to add product to shopping cart
Add Item to Cart	Cart_ID, Product_ID, Quantity, Unit_Price, Added_Timestamp, User_Session	Structured	Process data for adding item to cart
Remove Item from Cart	Cart_ID, Product_ID, Remove_Quantity, Action_Type, Removal_Timestamp	Structured	Process data for removing item from cart
Item Data	Product_ID, Product_Name, Price, Availability_Status, Product_Image, Description	Structured	Product information retrieved from inventory
Inventory Management System	Stock_Query, Product_ID, Available_Quantity, Price_Info, Product_Status	Structured	Communication with inventory system for product data
Retrieve Cart Data	Cart_ID, User_ID, Session_ID, Cart_Contents, Last_Modified	Structured	Request to retrieve existing cart information
Restore Cart State	Cart_ID, Previous_State, Cart_Items, Total_Amount, State_Timestamp	Structured	Data for restoring previous cart state
Validate Cart Contents	Cart_ID, Item_List, Validation_Rules, Stock_Check, Price_Verification	Structured	Cart validation and verification data
Shopping Cart Database	Cart_ID, User_ID, Cart_Items, Creation_Date, Last_Updated, Cart_Status	Structured	Complete shopping cart storage data
Shopping Cost	Cart_ID, Subtotal, Tax_Amount, Shipping_Cost, Discounts_Applied, Currency	Structured	Cart cost calculation details

Calculate Total Price	Item_Prices, Quantities, Tax_Rate, Discount_Codes, Shipping_Charges	Structured	Price calculation input data
Final Total	Cart_ID, Grand_Total, Tax_Breakdown, Applied_Discounts, Final_Amount	Structured	Complete cart total calculation result
Cart Summary	Cart_ID, Item_Count, Total_Amount, Cart_Items_Summary, Last_Updated	Structured	Summary of cart contents and totals
Cart for Checkout	Cart_ID, Validated_Items, Final_Total, User_ID, Checkout_Timestamp	Structured	Cart data prepared for checkout process
Order Processing System	Order_Request, Cart_Data, Customer_Info, Payment_Intent	Structured	Cart data transferred to order processing

Level 2: Customer Support System

Data Flow	Data Elements	Type	Description
FAQ Response	FAQ_ID, Question, Answer, Category, Keywords, Last_Updated	Structured	Frequently asked questions and their responses
Product Question	Question_ID, Product_ID, Customer_ID, Question_Text, Question_Category, Timestamp	Structured	Customer inquiries about specific products
FAQ Data	FAQ_Database_Query, Search_Terms, Category_Filter, FAQ_Results	Structured	FAQ database query and retrieval data
Plant Care Guides	Guide_ID, Plant_Type, Care_Instructions, Seasonal_Tips, Problem_Solutions	Structured	Comprehensive plant care guidance information

Plant Care Response	Response_ID, Customer_ID, Care_Instructions, Recommended_Products, Follow_up_Schedule	Structured	Personalized plant care advice and recommendations
Plant Care Query	Query_ID, Customer_ID, Plant_Species, Problem_Description, Care_Questions	Structured	Customer requests for plant care assistance
Plant Care Solution	Solution_ID, Plant_Problem, Diagnosis, Treatment_Steps, Product_Recommendations	Structured	Expert plant care solutions and treatment plans
Customer Care Team	Agent_ID, Expertise_Area, Available_Status, Current_Workload, Contact_Info	Structured	Customer service team availability and specializations
Support Ticket	Ticket_ID, Customer_ID, Issue_Type, Priority_Level, Creation_Date, Status	Structured	Customer support ticket creation and tracking
Support Ticket Data	Ticket_Details, Customer_Info, Issue_Description, Attachment_Files, Urgency_Level	Structured	Complete support ticket information and context
Technical Ticket	Ticket_ID, Technical_Category, System_Issue, Error_Logs, Severity_Level	Structured	Technical support ticket for system-related issues
Technical Issue	Issue_ID, System_Component, Error_Description, Impact_Level, Reporter_Info	Structured	Technical problem reporting and documentation

Technical Support Response	Response_ID, Solution_Steps, Resolution_Time, Escalation_Status, Follow_up_Required	Structured	Technical support team response and resolution
Technical Solution	Solution_ID, Issue_Resolution, Implementation_Steps, Testing_Results, Documentation	Structured	Technical problem resolution and implementation guide
Technical Request	Request_ID, Service_Type, Technical_Specifications, Priority_Level, Requestor_Info	Structured	Technical service requests and requirements

Level 2: Other Management Systems

Data Flow	Data Elements	Type	Description
Content Data	Content_ID, Content_Type, Title, Body_Text, Media_Files, Publication_Status	Structured	Website content and media information
Content Reports	Report_ID, Content_Analytics, Page_Views, Engagement_Metrics, Performance_Data	Structured	Content performance and analytics reports
Content Updates	Content_ID, Update_Type, Modified_Content, Editor_ID, Update_Timestamp	Structured	Administrative content modification data
User Data	User_ID, Profile_Info, Account_Details, Preferences, Contact_Information	Structured	Customer account and profile information
Account Reports	Report_ID, User_Analytics, Account_Activity, Registration_Stats, User_Metrics	Structured	User account analytics and reporting data

Account Actions	Action_Type, User_ID, Account_Changes, Admin_ID, Action_Timestamp	Structured	Administrative account management actions
Order Data	Order_ID, Order_Details, Order_Status, Customer_Info, Order_History	Structured	Complete order information and tracking data
Order Status Updates	Order_ID, Status_Change, Update_Reason, Updated_By, Status_Timestamp	Structured	Order status modification and tracking updates
Order Reports	Report_ID, Order_Analytics, Sales_Data, Revenue_Metrics, Performance_Stats	Structured	Order processing analytics and business reports
Order Notifications	Notification_ID, Order_ID, Customer_ID, Notification_Type, Message_Content	Structured	Order status notifications sent to customers

Level 3: JWT Token Generation Process

Data Flow	Data Elements	Type	Description
User ID + Device Info	User_ID, Device_ID, IP_Address, User_Agent	Structured	Verified user context from OTP process
Fetch User Details	User_ID, Device_ID	Structured	Request for user information from database
User Role & Permissions	User_ID, User_Role, Permissions, Account_Status, Full_Name	Structured	User authorization and profile data
JWT Payload	User_ID, User_Role, Permissions, Device_ID, Issue_Time, Expiry_Time	Structured	Token payload data for signing

Get Secret Key	Config_ID, Algorithm, Key_Request_Type	Structured	Request for JWT signing configuration
Secret Key Response	Secret_Key, Algorithm, Token_Expiry, Refresh_Expiry, Issuer, Audience	Structured	JWT configuration and signing key
User Context	User_ID, Device_ID, Session_Context, IP_Address	Structured	User session information for refresh token
Signed JWT	Access_Token, Token_Type, Algorithm, Signature	Structured	Complete signed JWT access token
Refresh Token	Refresh_Token, Token_ID, Expiry_Time, User_ID	Structured	Generated refresh token data
Token Pair	Access_Token_Hash, Refresh_Token_Hash, Token_ID, User_ID, Device_ID	Structured	Token pair for storage
Store Token Pair	Token_ID, User_ID, Access_Token_Hash, Refresh_Token_Hash, Issue_Timestamp, Expiry_Data	Structured	Complete token storage record
Log Token Issue	User_ID, Token_ID, Action_Type, IP_Address, User_Agent, Timestamp, Success_Status	Structured	Audit log entry for token issuance
Token Metadata	Token_ID, Issue_Time, Expiry_Time, Token_Type, Device_ID	Structured	Token information for response
Access Token	Access_Token, Token_Type, Expires_In, Scope	Structured	JWT access token for client
Complete JWT Response	Access_Token, Refresh_Token, Token_Type, Expires_In, Refresh_Expires_In, User_Info	Structured	Complete authentication response package

Token Generation Failed	Error_Code, Error_Message, User_ID, Timestamp, Failure_Reason	Structured	Error response for failed token generation
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Level 3: Device Session Management

Data Flow	Data Elements	Type	Description
Login Credentials + Device Info	Username, Password, Device_Fingerprint, IP_Address, User_Agent	Structured	User login attempt data
Validated User ID	User_ID, Authentication_Status, Permission_Level	Structured	Authenticated user information
Active Session Data	Session_List, Device_IDs, Status, Timestamps	Structured	Current user sessions
Previous Session Info	Old_Session_ID, Device_ID, Termination_Required	Structured	Session to be terminated
Force Logout Signal	Session_ID, Termination_Reason, Timestamp	Structured	Logout command to device
Login Success + Session Token	Session-Token, Expiry_Time, User_Permissions	Structured	Successful login response
Security Alert	Alert_Type, User_Info, Device_Details, Timestamp	Structured	Security notification data

Level 3: Search and Filter Processing

Data Flow	Data Elements	Type	Description
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Search Query + Filters	Search_Terms, Price_Range, Category, Size, Light_Req	Structured	User search input
Filter Configuration	Filter_Name, Options, Type, Display_Order, Status	Structured	Admin filter setup
Parsed Query	Clean_Terms, Operators, Modifiers	Structured	Processed search terms
Search Terms	Keywords, Synonyms, Weights	Structured	Search algorithm input
Available Filters	Filter_List, Current_Values, Constraints	Structured	Filter options for UI
Filter Parameters	Selected_Filters, Value_Ranges, Logic_Operators	Structured	Applied filter settings
Search Results Display	Plant_List, Total_Count, Page_Info, Sort_Order	Structured	Formatted search results

Level 3: Cart Contents Validation Process

Data Flow	Data Elements	Type	Description
Validation Request	Customer_ID, Cart_ID, Validation_Type, Request_Timestamp	Structured	Customer request to validate cart contents
Cart Request	Customer_ID, Cart_ID, Session_ID	Structured	Request to retrieve specific cart data
Cart Query	Customer_ID, Cart_ID, Status_Filter	Structured	Database query for cart items
Cart Items	Cart_ID, Product_ID, Product_Name, Quantity, Unit_Price, Line_Total, Item_Status	Structured	Retrieved cart item details

Stock Query	Product_ID, Requested_Quantity, Location_ID	Structured	Inventory availability check request
Stock Info	Product_ID, Stock_Quantity, Available_Quantity, Stock_Status, Restock_Date	Structured	Current inventory information
Quantity Check	Product_ID, Requested_Quantity, Available_Quantity, Max_Order_Quantity	Structured	Quantity validation data
Availability Results	Product_ID, Validation_Status, Available_Quantity, Issues, Recommendations	Structured	Stock availability results
Price Query	Product_ID, Customer_Tier, Quantity, Current_Date	Structured	Current pricing information request
Price Changes	Product_ID, Current_Price, Previous_Price, Change_Amount, Change_Date	Structured	Price modification details
Cart Updates	Cart_ID, Product_ID, New_Quantity, New_Price, Update_Reason, Timestamp	Structured	Cart modification data
Updated Items	Cart_ID, Modified_Items, Update_Summary, Total_Changes	Structured	Summary of cart modifications
Stock Issues	Product_ID, Issue_Type, Quantity_Short, Alternative_Available, Resolution	Structured	Out-of-stock problem details
Validation Data	Cart_ID, Total_Items, Items_Valid, Items_Invalid, Issues_Summary, Recommendations	Structured	Complete validation results
Notification Request	Customer_ID, Notification_Type, Message_Data, Delivery_Method, Priority	Structured	Notification generation request

Notification Data	Notification_ID, Customer_ID, Message_Title, Message_Body, Delivery_Method, Status	Structured	Complete notification record
Email Notification	To_Email, Subject, Body_HTML, Body_Text, Attachments, Priority	Structured	Email service delivery data
Activity Log	Customer_ID, Cart_ID, Action_Type, Details, Result, Timestamp, IP_Address	Structured	Audit trail information
Log Entry	Log_ID, Customer_ID, Validation_Type, Result, Issues_Found, Actions_Taken, Timestamp	Structured	Complete log record
Alternative Request	Product_ID, Customer_ID, Price_Range, Category_Preference, Stock_Required	Structured	Request for product alternatives
Alternative Query	Product_ID, Category_ID, Price_Min, Price_Max, Stock_Min, Similarity_Threshold	Structured	Database query for alternatives
Calculation Request	Cart_ID, Updated_Items, Discount_Codes, Tax_Rate, Shipping_Method	Structured	Total recalculation request
Validation Results	Cart_ID, Total_Amount, Items_Count, Changes_Made, Issues_Resolved, Alternatives_Suggested	Structured	Final validation outcome
Alternative Suggestions	Original_Product_ID, Alternatives_List, Comparison_Data, Stock_Status, Pricing_Info	Structured	Product alternative recommendations

Level 3: Order Tracking Process

Data Flow	Data Elements	Type	Description
Order Request	Customer_ID, Order_Number, Tracking_Request_Type, Session_ID	Structured	Customer request for order information
Order Query	Customer_ID, Order_ID, Order_Number, Date_Range	Structured	Database query for order details
Status Request	Order_ID, Status_Type, Last_Update_Time	Structured	Request for current order status
Order Details	Order_ID, Order_Number, Customer_ID, Total_Amount, Order_Status, Tracking_Number, Items_List	Structured	Complete order information
Order Status Update	Order_ID, New_Status, Status_Date, Location, Carrier, Notes, Updated_By	Structured	Status change information from administrator
Status Change	Status_ID, Order_ID, Status_Code, Status_Description, Location, Timestamp, Carrier_Info	Structured	Updated status information
Status Report	Order_ID, Current_Status, Status_History, Timeline, Delivery_Estimate, Exception_Flags	Structured	Complete status report for administrator
Notification Data	Customer_ID, Order_ID, Notification_Type, Message_Template, Status_Info, Delivery_Preferences	Structured	Data for generating notifications
Email Notification	To_Email, From_Email, Subject, Body_HTML, Body_Text, Order_Attachments, Tracking_Links	Structured	Email notification to external service